The results are shown in Figure 10. Even though the model did not see the +2 and +3 tasks, it appears that prefix-tuning can generalize from the +1 task. The +1 + LessThan task is another example of successful prefix-tuning for a task that is compositional in pretraining tasks. Similarly, for Divisible +1, LessThan +1, LessThan + $\nearrow$, MoreThanEqual, NotDivisible,

Similarly to the case in Appendix D.1, we observe several instances in which the largest, 32-layer, model performs worse, when prefix-tuned, than the smaller models, as well as cases where shorter prefixes perform better than longer ones.

E FURTHER EXPERIMENTS WITH MEMORIZATION

Our experiments in Section 5 focused on algorithmic tasks such as sorting, incrementing and counting. However, learning a natural language also includes a substantial memorization component. For example, learning a new language requires learning its vocabulary. Hence, if a novel task requires memorization of novel concepts, the fine-tuning method should be able to memorize them. To this end, we evaluate and compare the abilities of prefix-tuning and LoRA to learn to memorize a large number of new words and show that, for the same amount of learnable parameters, LoRA can learn to translate to a new language while prefix-tuning cannot. These findings further strengthen our results from Section 5.

We use the PHOR-in-One dataset (Costa et al., 2022) that contains 4921 unique translations of English words into German, Spanish and Portuguese. The dataset is preprocessed to remove all accents in order to ensure that fine-tuning does not require characters that the model has not seen during pre-training (e.g., *éçü* would become *ecu*). Character-level tokenization is used, with the additional `<TR>` token that separates the source and target words and a `<PAD>` token that we use to ensure all training sequences are of the same length.

We then pre-train a 4-layer 4-head transformer to translate the English words to German. As seen in Table 4, this model successfully memorizes 99.3% of the English-German pairs. We then want to fine-tune this model to instead translate the English words to Spanish. Spanish is linguistically closer to English than German, so, in a way, the fine-tuning task is simpler than the pre-training task.

We do prefix-tuning on English-Spanish pairs with a prefix of size $n_S = 66$ but it achieves only 0.18% accuracy (about 10 words which are the same in English and Spanish, e.g., *instrumental*, *orbital*, *solar*). However, rank-4 LoRA is able to achieve 94.8% accuracy with half the training iterations. Both fine-tuning methods have similar numbers of learnable parameters (see Table 4). Therefore, this is further evidence that prefix-tuning fails to learn a new task, while LoRA with the same number of learnable parameters can.

Note that there is no train-test split for the word pairs. We are evaluating the accuracy on the training set as this experiment is measuring *memorization* rather than *generalization*.

Table 4: Further experiments on memorization of word pairs form (Costa et al., 2022) in different languages. The model is pre-trained on the English-to-German word pairs to almost perfect accuracy. Prefix-tuning fails to modify it to memorize English-Spanish word pairs while LoRA with the same number of parameters and half the training iterations reaches 94% percent accuracy. 5 random seeds.

	Pre-training on English to German	Prefix-tune (n_S=66) on English to Spanish	Rank-4 LoRA on English to Spanish
Accuracy	99.3 ± 0.1%	0.18 ± 0.01%	94.8 ± 1.1%
Learnable parameters	3.19M	67 584	66 780
Training iterations	50 000	100 000	50 000